

## Problem

Design a problem to help other students better understand low-pass filters described by transfer functions.

## Step-by-step solution

## Step 1 of 1

Design a problem to help other students better understand lowpass filters described by transfer functions.

Although there are many ways to solve this problem, this is an example based on the same kind of problem asked in the third edition.

**Problem**

Determine the cutoff frequency of the lowpass filter described by

$$H(\omega) = \frac{4}{2 + j\omega 10}$$

Find the gain in dB and phase of  $H(\omega)$  at  $\omega = 2$  rad/s.

**Solution**

$$\text{At dc, } H(0) = \frac{4}{2} = 2.$$

$$\text{Hence, } |H(\omega)| = \frac{1}{\sqrt{2}} H(0) = \frac{2}{\sqrt{2}}$$

$$\frac{2}{\sqrt{2}} = \frac{4}{\sqrt{4 + 100\omega_c^2}}$$

$$4 + 100\omega_c^2 = 8 \longrightarrow \omega_c = 0.2$$

$$H(2) = \frac{4}{2 + j20} = \frac{2}{1 + j10}$$

$$|H(2)| = \frac{2}{\sqrt{101}} = 0.199$$

$$\text{In dB, } 20 \log_{10} |H(2)| = -14.023$$

$$\arg H(2) = -\tan^{-1} 10 = -84.3^\circ \text{ or } \omega_c = 1.4713 \text{ rad/sec.}$$